The agent starts like an agent and stops like a coin flip

A computational-neuroscience dissection of why humans solve ARC-AGI-3 and an LLM agent does not

Draft v2, 2026-10-22. Every figure and number here was measured by us from artefacts in the linked repository, with one script per claim (Appendix B). Figures from other people's work are quoted and attributed. Where something is unverified, the text says so.

Abstract

ARC-AGI-3 scores Relative Human Action Efficiency: how few actions an agent needs to clear the levels of games it has never seen, against a human baseline. Humans solve 100% of its games; the open LLM agents that win its public milestones clear a few levels per game. We ask the question the way a systems neuroscientist would: decompose the human solution into faculties, measure each one in the agent against ground truth, intervene on each, and see what moves. Three ground truths make this possible — 6,512 recorded human actions on 22 public games (168 level completions), random and clairvoyant policies under the same budgets, and the game engine itself — on the two open harnesses that won the competition's milestones (Qwen3.8-27B; Qwen3.8-Flash-Next, 180B).

The ledger that results is the paper. *Contingency detection* is near-human (94% of actions change the board; humans 98.7%). *Action efficiency* is super-human: over 775 cleared levels the agent used a median 0.67 of the human baseline and its efficiency term sits at 1.05 against a cap of 1.15, so the score is depth and nothing else. *Object and agent tracking* is partial: a theory fitted by counting — the thing that moved last is the agent, it translates per key unless a colour that blocked before is in the way — predicts 72% of the direction transitions humans produced, and 46% when the agent is treated as a colour rather than an object; in 40% of the public games no single object ever moves. *Dynamics induction* is slow: what a counter predicts, the model re-derives in six to ten turns per level. *Prediction-error learning*, *replay*, and four further brain-inspired protocols were adopted by the model and changed nothing. *Objective inference* is the deficit: the colour the agent enters at a level-completing action is the game's modal colour in 76% of completions, and a planner aimed at it in the fitted theory, re-planning on surprise in the real engine, clears 0 of 37 levels — on dc22 it reaches the goal in 20 actions where the human used 55 and nothing happens, because objectives are multi-step causal chains and the colour is their signature. An open model asked to author the abstraction that encodes such chains succeeds on 4 of 22 games in 437 attempts; the strongest system on the benchmark buys it in context at $19,000–26,000 per game. *Effort allocation* — the one lever that acts solely in the hidden nine-hour run — is bounded by a calibrated simulator at +4 points over the winning harness's priority and +2 over the field's best retune; throughput, not allocation, binds.

The instrument is part of the finding. Each team displays its best draw, so the leaderboard's middle is a max-statistic: a fixed configuration drawn daily has standard deviation 2.7–3.5 at the 27-point level, 76 one-shot copies of the winning notebook scored mean 26.8 and maximum 36.1, and the retune advertised at 31–32 scores 27.2 per draw. Executing anything in the harness's customisation hook costs 0.448 of score; the engine's UNDO reverts the board in 1 of 25 games; the rendered frame is not a sufficient state until the animation frames every implementation discards are kept (repeat agreement 43% → 90%).

At Marr's computational level the task is joint inference of dynamics and objective under core-knowledge priors; at the algorithmic level, the local parts (agent, dynamics, goal signature) are cheap and the objective's causal chain is a structured search the model performs in context, slowly, and no module we attached could perform for it; at the implementational level the brain pays with dedicated systems and the agent pays in generated tokens on one GPU, which makes allocation and throughput the whole economics of a run. We close with a theory-based harness in which code owns what counters already predict and every token goes to the chain, and three falsifiable predictions.

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1. Introduction: the question at the computational level

Chollet defines the intelligence of a system as "a measure of its skill-acquisition efficiency over a scope of tasks, with respect to priors, experience, and generalization difficulty" (Chollet 2019). ARC-AGI-3 makes that definition interactive: the score is Relative Human Action Efficiency, the agent's per-level action count against "the second-best human action count" (ARC-AGI-3 Technical Report, 2026). The benchmark constrains what knowledge is admissible: its environments "only leverage Core Knowledge priors", and "humans can solve 100% of the environments, in contrast to frontier AI systems which, as of March 2026, score below 1%". That constraint is inherited from Chollet, who prescribes that a test "should only feature tasks that assume the four core knowledge systems", and from Spelke & Kinzler (2007), whose taxonomy — objects, agents, number, geometry — Chollet is naming.

So the benchmark is a cognitive-science artefact by construction, and the methods that do best on it are not. The nearest published work to ours in constraints, Rudakov, Shock & Cowley's training-free graph exploration, grounds itself in Go-Explore, MuZero and curiosity; the field's winning open harnesses are engineering reports. This paper does the arithmetic the other way round. Humans solve these games because they bring a structured hypothesis space about objects and agents and infer, from one or two interactions, a game's dynamics and its objective — theory-based reinforcement learning in Tsividis et al.'s (2021) sense, the account of why people learn an Atari game in minutes. We ask which parts of that solution an LLM agent has, by measuring each part against the human data the benchmark was calibrated on and a random floor, and by intervening on each part in the harness and reading what moves.

The result is a ledger (§3) whose rows are faculties and whose entries are measurements, and a conclusion that no single intervention could have reached: perception, motion and efficiency are not the gap, allocation is a small one, and the deficit is the inference of an objective's causal chain — the same capability a forward experiment on abstraction authoring locates, and the one the strongest frontier

system buys in context at a price. Along the way the instrument itself had to be characterised: a leaderboard that displays each team's best draw, a measurement channel with a hidden tax, an observation that is not a state, an engine whose UNDO is a no-op. §2 does that first, because nothing after it can be read without it.

1.1 Prior work

The benchmark and its priors. Chollet (2019) defines intelligence as skill-acquisition efficiency under priors and frames the core-knowledge systems a test may assume; the ARC-AGI-3 Technical Report (2026) instantiates them interactively as Relative Human Action Efficiency against a human baseline. Spelke & Kinzler (2007) give the object/agent distinction this paper measures; Dubey et al. (2018) show by masking that the object prior is the one whose removal costs human players most. Lake et al. (2017) argue for causal world models, compositionality and learning-to-learn as the ingredients of human-like learning; Tsividis et al. (2021) show that an agent that infers a game's theory and plans in it learns Atari games at human speed — the account whose parts our ledger tests one at a time.

Methods on ARC-AGI-3. The open harnesses that won the competition's milestones (Tufa Labs' Duck harness, June; Franzen's patched Duck harness with Qwen3.8-Flash-Next, September) are LLM tool-use loops with a Python sandbox; we instrument both. Rudakov, Shock & Cowley (2025) reach a strong training-free result with graph-based state-space exploration; Rodionov (2026) induces executable world models verified by replay with frontier models, which §4.3 tests the precondition of. ARC Prize's analysis of GPT-6 Astra (2026) documents in-context symbolic world modelling at $19,000–26,000 per game, and OpenAI's provider-adapter results (2026) document a harness effect of thirty-seven points on one model — the confound §2.2 measures at the kernel level. Affectify's published slot-priority retune (2026) is the field's discovery of the allocation lever §7.2 bounds.

Brain-inspired mechanisms we tested. Predictive coding (Rao & Ballard 1999; Friston 2010) as a predict-then-check loop; prioritised replay (Mattar & Daw 2018) as harness-driven replay of retained solvers; model-based control with arbitration (Daw et al. 2011) as a gated planner; directed exploration as an orienting census; resource-rational effort allocation (Lieder & Griffiths 2020; Shenhav et al. 2013) as the scheduler's priority. Each appears in §6 or §7 with its measurement.

2. Approach I: instruments — three ground truths and two confounds

2.1 Chassis, ground truths, and the two contracts

Two open harnesses were instrumented. The Milestone-1 winner (Tufa Labs' Duck harness, Qwen3.8-27B-FP8 through vLLM) served the measurements of §4–§5 on the 25 public games at a per-game cap of 7,920 s. The Milestone-2 winner (Franzen's patched Duck harness, Qwen3.8-Flash-Next (a 180B-parameter mixture of experts; Hugging Face safetensors total 1.80×10^{11}) in Intel's AutoRound W4A16 quantisation with an MTP drafter on the Pennyroyal SGLang fork) served the October measurements, our leaderboard draws, and the allocation study of §7. Both run offline on one RTX PRO 6000.

The linked submission. The Kaggle entry this paper is linked to is `tantan0327/arc3-franzen-dprime`: Franzen's Milestone-2 notebook, byte for byte (Tufa's Duck harness under MIT, his Apache-2.0 patch, Qwen3.8-Flash-Next in Intel's AutoRound W4A16 with albucino's INT4 MTP drafter under the Qwen Community licence, the Pennyroyal SGLang fork under Apache-2.0), plus one cell of ours: a clean-room, CC0 implementation of the published D' slot priority (`scripts/patches/priority_form.py`, §7.2). Its daily draws are the D' series of §2.3; its best public draw is 33.55 and its per-draw mean 27.2. Everything else in this paper is measurement around that entry and its predecessors, not part of it.

Three ground truths. **Human replays:** the official human solutions of 22 public games — 6,512 actions, 168 level completions — replay 22/22 in sync on the local engine against reference solutions whose continuous integration asserts a score of exactly 100.0. **Policy floors:** a random policy on the same 25 games with twice the summed human baseline as its budget, and a clairvoyant policy where a bound is needed (§7). **The engine:** every claim about transitions is checked by executing actions, not by reading frames.

Two contracts. The author's 25-minute, ten-game demonstration gives each game its own stream, so nothing competes for the model; the hidden nine-hour run plays ~110 games on ten model slots. Everything measured on the demonstration is a per-game measurement; only the board can see allocation (§7).

2.2 The channel: executing anything in the customisation hook costs 0.448

The Milestone-1 harness offers a customisation hook for exactly the kind of modification an experiment needs. **Executing anything in it costs 0.448 of leaderboard score.**

```
unmodified draws          0.95  0.98  1.11  1.14  1.20  1.23
hook arms                 0.58  0.60  0.69  0.72  0.74
                                ^ the null control
```

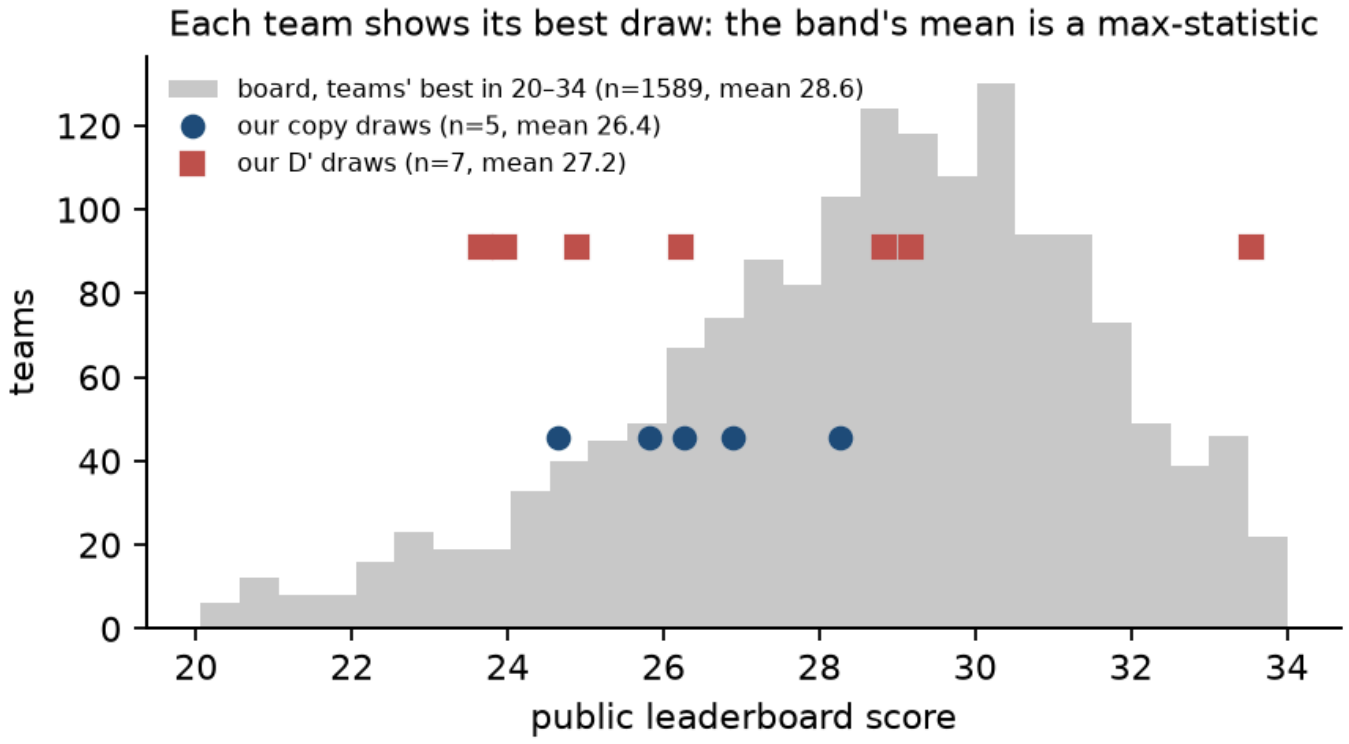
The null control imports a module already in `sys.modules` and prints its length. It drew 0.60, and on replication 0.59; six against six with no overlap, exact permutation p = 0.0079. We could not find the mechanism — throughput is unaffected, cold-starting is refuted, the module does no work — and we report it for what it does to other people's results: any modification shipped through that hook must be worth 45% of the score to break even, and we cannot know how many shared results on this harness were measured through it. Our own measurements use a substitution on the harness's setup commands, verified clean (a null substitution scored 1.20). The effect is not a quirk of one notebook: the frontier labs have since reported the same confound at scale — GPT-6 Astra scores 62.7% through a provider-neutral interface and 99.9% through its own adapter, whose only additions are retained notes and compacted reasoning (OpenAI 2026; confirmed by the ARC Prize Foundation). A leaderboard number on this benchmark is a joint measurement of a model and the software around it.

2.3 The leaderboard is a max-statistic, and it manufactures a wave

The board keeps each team's best submission, and most entrants fork a few public recipes. That makes its upper band a max-sampling artefact, and we can show it with our own draws. At the 1-point scale two fixed configurations drawn four times each gave sd 0.10 and 0.20; under that distribution six hundred

teams keeping the best of about five draws should put 67 teams above 2.0, and 66 were observed. We then ran the experiment: the top public kernel, advertising 2.23, forked byte for byte and submitted, scored 1.22. Six draws of one unchanged recipe later climbed 2.45 → 3.92 and carried the same bytes from ~200th to ~77th place.

The field then ran the experiment for us at the 27-point scale. When the Milestone-2 first place published his notebook, hundreds of teams submitted one copy each: 76 one-shot copies scored **mean 26.8, sd 3.55, maximum 36.1** against the author's own best of 27.9, and the top-100 line moved from 4.9 to 28.2 in two days. When one team published a retuned slot priority ("D'", §7.2) its single submission showed 31.54 and three copies 30.8–31.9, all displayed as bests; our daily draws of a clean-room implementation of the same formula scored 23.95, 23.66, 28.83, 33.55, 29.18, 26.23, 24.88, 25.55, 29.35 — **mean 27.2** — against our byte-identical copy's 26.3. The board's "copy band" (1,661 teams between 20 and 34 on 2026-10-20) has mean 28.7 for the same reason: per-draw means sit about two points below what the board displays (Fig. 1). Single-draw comparisons measure nothing; "best submission" among copies ranks luck. The top of the board is another matter — the leading teams sit tens of standard deviations above any public family — and that is exactly what the public band is not.



2.4 Noise, and the decision rules it forces

A fixed configuration drawn daily on the board has sd 0.47 at 3 points and 2.7–3.5 at 27; the offline instrument (25 games, full length) has sd ≈ 3 levels on 32–45 with per-game ranges wider than any single-change effect we measured (lp85 2–6 levels across three runs of one notebook). An independent team reported the same instrument in the forum: two runs of an identical build, back to back, 2.60 and 1.45. A three-draw family mean on the board has standard error about 0.3 at the low scale and 1.6 at the 27-point scale; it resolves a difference of one or two points and nothing smaller. Every family decision in

this paper was therefore pre-registered as a three-draw mean against a threshold; offline totals were never used to decide anything, because a total is a sum of per-game draws whose individual ranges exceed any effect we measured; the only offline signal treated as readable was a per-game flip outside that game's own range with no game falling below its range, and an arm that moved a total without moving any game out of its range was read as noise. Under that rule, of thirty-odd single-change arms measured at full length, one was weakly positive. Three of twelve competition reruns in October failed at dataset mount and scored 0; zero-score draws are excluded from every family mean. Appendix C lists every draw.

3. Approach II: the dissection, as a faculty ledger

The human solution decomposes into faculties that cognitive science and systems neuroscience name and, in several cases, localise. Each row below is a faculty, the model that describes it, the measurement we made of it in the agent, and the verdict; the section that carries the measurement is named.

human faculty	brain system / computational model	measurement	verdict	§
contingency detection: what my action affects	agent-detection signature (Spelke & Kinzler)	94% of agent actions change the board; humans 98.7%	human-level	4.2
action efficiency	the agent signature "efficient means"	775 cleared levels: median actions/baseline 0.67; efficiency term 1.05 vs cap 1.15; 84% at ≥1.0	super-human	4.2
object permanence; the agent as an object	ventral stream; core-knowledge object system	code tracks the agent through 72% of human direction moves as an object, 46% as a colour; no single object moves in 40% of games	partial	4.4, 5.1
sufficient state (perception)	—	repeat agreement 43% as rendered, 90% with the animation frames the harness discards	repairable	4.3
dynamics induction (theory of the mechanics)	theory-based RL (Tsividis); model-based learning	a counter predicts 72% of transitions; the model re-derives mechanics in 6–10 turns per level	slow	5.1
prediction-error learning	predictive coding (Rao & Ballard; Friston)	predict-then-check loop adopted (474 hits / 140 misses); levels unchanged	null	6
consolidation and replay	hippocampal replay (Mattar & Daw)	retained solver replayed by the harness: 2 of 45 replays cleared a level	null	6
directed exploration	novelty / information gain	an orienting census (not read: platform failures); four prompt-level arms null	null / untested	6
objective inference	OFC/vmPFC goal representation; structure learning	76% of completions enter the modal colour; a planner on that goal clears 0 of 37 levels; an open model authors the abstraction on 4 of 22 games	**the deficit**	5.2–5.4
effort allocation (metacontrol)	ACC; resource rationality (Lieder & Griffiths; Shenhav)	calibrated simulator: oracle +4 over base, +2 over the field's retune; +1 per draw measured	small	7
measurement	psychometrics	best-of-draws board; per-draw sd 2.7–3.5; hook tax 0.448	method	2

The brain mappings organise the measurements; they are not themselves measured, and §10 says where a row has no evidence behind its mapping. What the ledger shows at a glance is the shape of the result: the bottom-up faculties are present or repairable, the interventions that imitate learning mechanisms are null, and the one empty row is the top-down one.

4. Results I — perception and action: what the agent has

Sections 4–7 are the results, in the ledger's order: bottom-up faculties first (§4), the dynamics and the objective (§5), the interventions (§6), the economics of the run (§7). Leaderboard numbers are in §2.3 and Appendix C; every offline number is a public-game measurement.

4.1 Starts like an agent, stops like a coin flip

Against the random floor, on the same games and the same budget:

```
clears at least one level     agent 11 / 25  (44%)     random 2 / 25  (8%)
finishes the game             agent  0 / 25            random 0 / 25
```

The agent is 5.5× chance at getting started and exactly equal to chance at finishing. Whatever it has, it spends in the opening. The per-level action counts say where it goes: a level the agent completed took a median of 20 actions and never more than 150; a level it stalled on burned a median of 122 and up to 397. **The agent does not fail by being slow; it fails by continuing.** The stall has a signature the harness could read online: within the agent's own stream, completed levels run at ~0.8 bits of action-type entropy while stalled tails hold 1.0–1.13, and the chrome-masked novelty of the boards it reaches climbs through completed levels (85.8% → 97.5%) and collapses through stalled ones (90.8% → 78.5%). Humans run at 98.4–99.6% novelty and rising to the last action of every level: nearly every human action moves the world somewhere new, which is close to a policy invariant on these games. The 60-minute traces of §7 add the other half: the games that stall in 25 minutes are slow, not stuck — given an hour each, eighteen of them cleared 69 levels.

4.2 Contingency, and efficiency at the cap

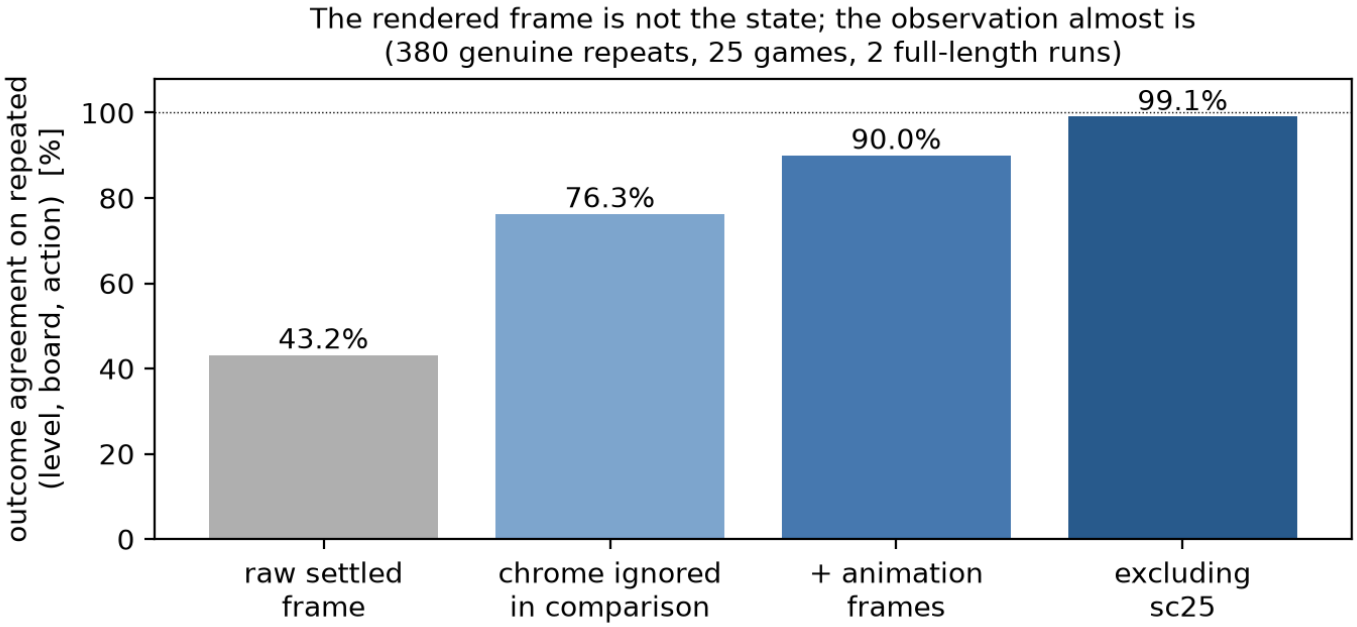
Spelke lists contingency and efficiency as signatures of agent representation, and the agent has both. 94% of its actions change the board, against a human 98.7%; true cycling is 2.3% and uncorrelated with score; the 13.9% of actions that leave the board unchanged (human 0.6%) are not the gap, because removing them does not follow through to levels. On the 180B chassis we compared, for every one of 775 cleared levels, the agent's actions with the human baseline the scorer uses: the median ratio is 0.67 (quartiles 0.50–0.90), the mean efficiency term 1.05 against the 1.15 cap, 84% of cleared levels at or above 1.0. Holding levels fixed and raising every term to the cap adds 8%; raising them to 1.0 would lower the score. The agent is more action-efficient than the humans it is measured against on the levels it clears; everything it fails to score is a level it never reached. **For this class of agent the metric is depth, and nothing else.** (The scorer caps at 1.15 where the published formula says 1.0; Appendix A.)

4.3 The observation is not a sufficient state

The strongest published results on ARC-AGI-3 induce an executable world model — the agent writes code that predicts the next board — and verify it by exact replay against observed transitions (Rodionov 2026). That rests on an assumption the papers state and do not test: that the rendered frame is a sufficient state. Every recorded transition is a (level, rendered board, action) → board triple; collisions on the first three with disagreement on the fourth are proof of state the frame does not show.

```
380 repeats of an identical (level, rendered board, action)
  as rendered                          216 contradictions    43.2% agree
  ignoring chrome when comparing        90                   76.3% agree
  + conditioning on the animation frames  38                 90.0% agree
```

The failure is rendered instrumentation, not physics: counters on row 0, a bar draining along row 53. They are found not by volatility but by **isolation** — a cell is instrumentation if it changes while nothing else does. The second repair costs nothing: the harness defines `frame` as the final visible frame and `animation_frames` as the rest, and every implementation we know of discards them by taking `frame[-1]`. Sixteen of the eighteen games with repeats are then perfectly deterministic; lf52's 49 contradictions are all distinguishable in the preceding animation frames; sc25 alone holds 35 of the 38 unexplained cases, and excluding it the agreement is 99.1% (Fig. 2). A correction we owe the reader: our first version reported 93.4% by masking the board before keying, which made distinct states collide; the honest chrome-masked figure is 76%.



4.4 Objects and agents

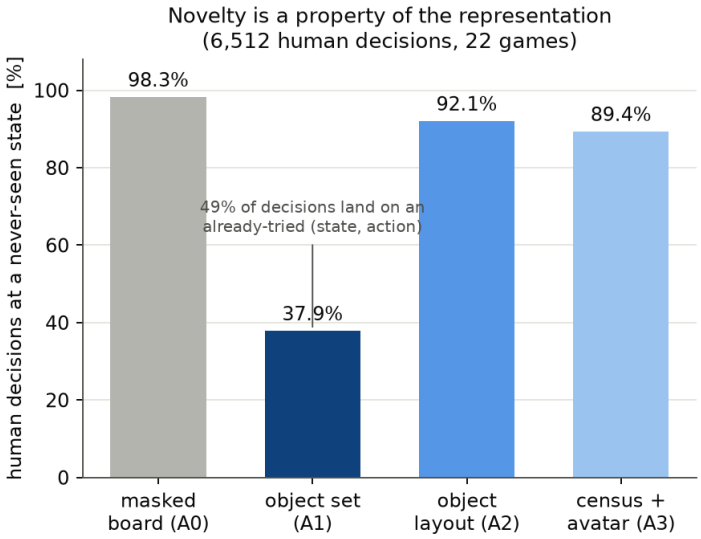
Chollet's core knowledge systems separate objectness — "cohesion, persistence, and contact" — from agentness and goal-directedness, and Spelke & Kinzler say why: "Spatio-temporal principles do not govern infants' representations of agents, who need not be cohesive, continuous in their paths of motion, or subject to contact … Instead, the intentional actions of agents are directed to goals, and agents achieve their goals through means that are efficient." The entity an ARC-AGI-3 player controls is an agent; implementations treat it as an object a direction key translates.

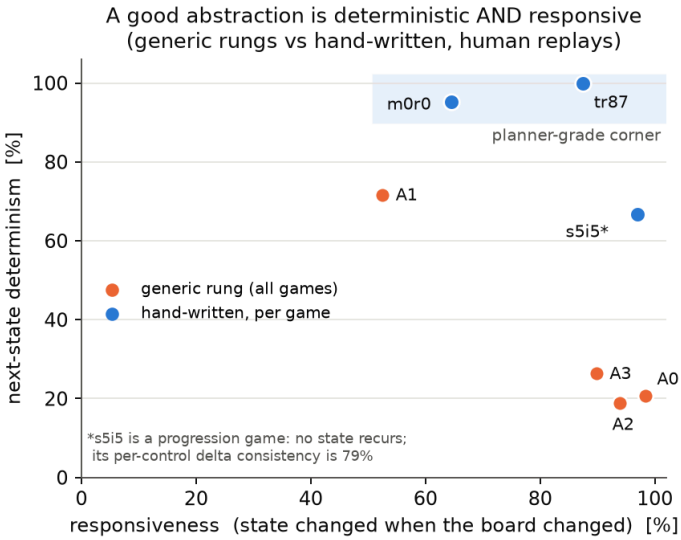
We measured whether that assumption holds (`scripts/measure_agency.py`): for every direction press that changed the board, does the change read as one connected object translating? Across three recorded runs on three harness generations the classification is stable per game: one object accounts for ≥80% of translating moves in 3–5 of 17 games, partially in 6–8, and never in 6–8 (cn04, tu93, lf52, cd82, re86, ls20; ar25 ≤1%). In roughly 40% of the public set no direction press ever reads as an object moving; the avatar the engineering practice assumes is reliably present in under a fifth. §5.1 measures the same distinction a second way: a theory that tracks the agent as an *object* — the thing that moved last, whatever its colour — predicts 72% of human direction moves, and the same theory keyed on *colour* predicts 46% (re86 falls from 87% to 2%). RHAE itself, measuring action efficiency against a human, is a measurement of a core-knowledge agent signature — a stronger justification of the metric than the benchmark offers.

Supplying object identity to the agent as a goal vocabulary, finally, degrades it monotonically in the sharpness of the identity (colour 13/22 games, size class 12/22, exact shape 3/22) — the mirror image of Dubey et al.'s (2018) human result, where removing the concept of objects is the costliest ablation. The prior humans need most is one the agent does not use, and handing it over explicitly makes play worse.

4.5 Novelty is a property of the representation

A sufficient state settles whether a world model can be sound, not whether it is useful: on the 6,512 human decisions, 98.3% are taken at a masked board never seen before in that game (0.9% at a state-and-action pair already tried). At that representation there is nothing to plan over, which is why mechanical frontier navigation measured null (§6). But novelty is a property of the representation, not of the games. Re-measured up an abstraction ladder (`scripts/measure_abstraction_ladder.py`), a bare object census (colour, count) takes novelty to 37.9% with 49% of decisions at an already-tried state-action pair, while positional rungs keep the distinctions and stay ~90% novel. No fixed rung is the answer; the planner-grade corner is reached only by choosing which variables matter for *this* game: three hand-written, game-specific abstractions reach next-state determinism of 100%, 95% and a per-control delta consistency of 79% at revisit rates of 10–27%, where every generic rung is either blind or never repeats (Fig. 3). With those abstractions standing in for authored ones, a breadth-first planner over observed abstract edges re-solves 70 of 70 level segments of the human trajectories in 24% fewer actions than the humans. That is, to the letter, what the strongest agent on the benchmark does: ARC Prize's analysis of GPT-6 Astra shows "on-the-fly symbolic world modeling" in a self-invented shorthand — game-specific variable selections — at $19,000–26,000 per game against a $12.78 human baseline. The ladder locates the capability precisely: not search, not memory, not planning, but the authoring of the abstraction.





5. Results II — dynamics and objective: where the deficit is

5.1 Theory induction by code

If the agent's slowness is slow theory induction, the natural test is whether a code-fitted theory — no model, no tokens — predicts the transitions humans produced. We replayed the 22 human solutions on the local engine and fitted, online, the simplest theories Spelke's systems license (`scripts/test_theory_induction.py`). Theory C: the thing that moved last is the agent; under a direction key it translates by a fixed per-key offset unless the cells it would enter hold a colour that has blocked it before. Theory D adds what a second look at the failures demanded: a masked border (the timer bar changes every action), a per-key step length (ar25's sprite moves three cells per press) with a slide-until-blocked alternative chosen online by predictive score (g50t), blocking learned over the whole sweep, and the agent chosen by continuity first and contingency second when several things move. Each prediction is made before the transition is seen and scored exactly: moved or not, in which direction, how far (±1).

game	C	D	game	C	D
dc22	95%	95%	sp80	66%	64%
cn04	87%	**92%**	ar25	**61%**	39%
re86	87%	**91%**	bp35	48%	49%
g50t	34%	**77%**	tr87	38%	33%
lf52	**74%**	38%	sc25	33%	33%
ka59	70%	**74%**	tu93	28%	33%
wa30	69%	**72%**	sk48	69%	70%
m0r0	66%	68%			

Over 4,083 direction transitions, C predicts 68%, D 67%, the better of the two per game 72%. Click games are trivially regular (97–100% of clicked colours behave as before) and are not the question. The games the theories miss — ar25 and tu93 below 40%, bp35, sc25, tr87 — are the ones whose sprites deform or whose dynamics involve a second moving body: object identity under shape change and

multi-object interaction, what human vision supplies for free. Everything else about movement — translation, step, blocking, sliding — a counter learns from a handful of transitions, and the model re-learns at every level in six to ten turns of tokens.

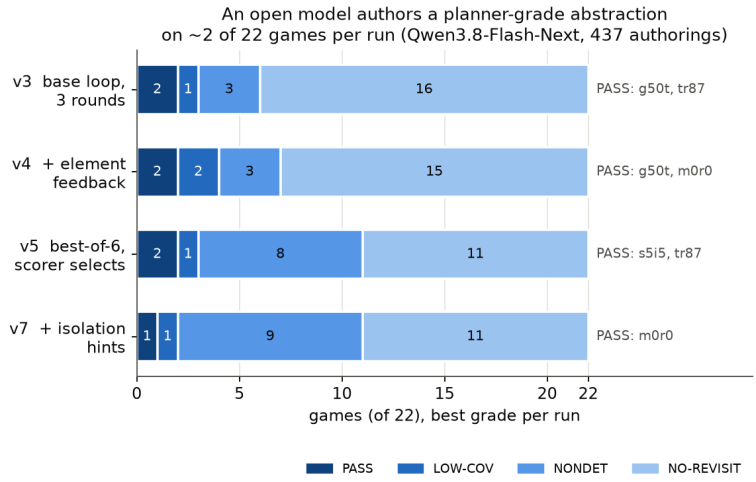
5.2 The goal is cheap; the objective is not

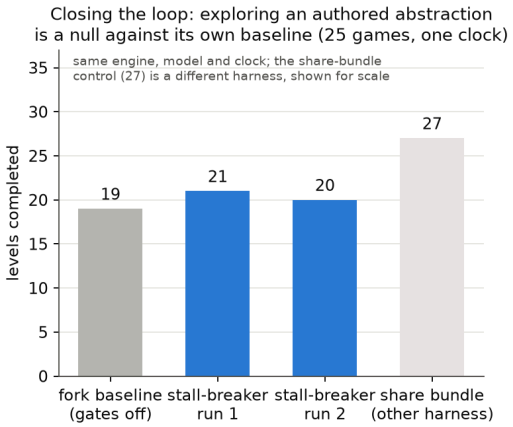
With the agent tracked, the goal is cheap to read: at the action that completed a level, the colour the agent entered or clicked is the game's modal colour in 76% of 129 completions, and in dc22, g50t, ka59, r11l, sk48, tn36 and vc33 in every one. "The level ends when you reach colour X" is inferable after one completion in most games.

Then the planner test says what the objective is not. We closed the loop (`scripts/test_theory_planner.py`): fit the theory on the human's level 1, take the colour entered at the completing action as the goal, and let a breadth-first planner in the fitted theory play levels 2 and up in the real engine — plan, act, compare each outcome with the prediction, re-plan on surprise, with the four direction keys given their named directions as a prior and the board's dominant colour excluded from goals. It completed **0 of 37 levels** across eleven direction games. The planner is not the problem: on dc22 it reached the goal colour in 20 actions where the human used 55, and nothing happened, because reaching that colour ends the level only after something else has been done. The public games' objectives are multi-step and dynamic — collect, then exit; toggle, then pass; push, then align — and a static "reach colour X" captures the last step of each. The 76% is a *signature* of the objective, not the objective.

5.3 Can the model author the abstraction?

We tested the capability the ladder names, directly. An open 180B model served locally was shown a few frames and transitions of each of the 22 replay games and asked to write `extract_state`; each result was scored against the human replays on next-state determinism and revised on concrete feedback. Over five runs and 437 authorings the model produced a planner-grade abstraction on **4 of 22 games** (determinism 92–100% at revisit rates of 13–36%, two of them matching our hand-written references). Feedback, best-of-6 selection with the scorer as selector, and handing the model the instrumentation cells our isolation rule finds each changed *how* the other games failed and none changed how many succeeded. The model writes perception code easily; on four games in five it does not choose the right variables, and the harness can supply sound parts but not the choice (Fig. 4). Wired into the harness as a stall-breaker — after a fixed number of actions on a level without completing it, the harness asks the model for an abstraction, scores it on the agent's own recorded transitions, and if one passes explores the abstract state space systematically in bursts of engine-speed actions, Go-Explore style, handing control back between bursts — the loop ran cleanly across 25 games (abstractions authored on 13, adopted on 5, 1,617 planned actions) and completed 21 and 20 levels against the fork's own gate-off baseline of 19 — a null. Offline, the same explorer clears a maze level in roughly 850 actions that a random policy never clears, and clears nothing on a tile puzzle or a progression game whose goals a human sees at a glance: systematic coverage substitutes for wandering, not for goal-directed play. A methods lesson is attached: our first comparison, against 27 and 28 on a different bundle, was confounded and withdrawn, and the bundle effect (19 vs 27–28 on the same engine and model) is larger than any modification we measured.





5.4 The deficit, located three ways

Three independent routes arrive at one place. Imitation of what humans did (two fine-tunes, the second on the harness's own rendering: 17 vs 20 and 3 vs 6 levels), retention of context, and an executable world model in both forms failed, the harness-driven form at the source: 98.3% of decisions on a solved trajectory occur at a board never seen before. A code-fitted theory predicts 72% of human movement and 76% of goal signatures and clears no level, because objectives are causal chains the theory space does not contain. And the model, asked to author such a space, succeeds on four games in twenty-two. What humans supply at 100% is the inference of that chain from one or two interactions; what the LLM supplies, slowly and at a price, is the same thing in tokens. Neither a counter nor a search substitutes for it, and — §6 — neither does any learning mechanism we attached.

6. Results III — interventions: the brain-inspired arms, and what moved

Each arm is one change on an otherwise unmodified harness, measured at full length against the vehicle's band (§2.4), with the mechanism it implements and the faculty it targets. The 27B arms ran on the Milestone-1 harness; the 180B arms on the Milestone-2 notebook, where the band is 44/50/40 levels on the ten-game demonstration and the one precise metric is per-stream throughput (62.6 tokens/s, sd 1.5 over fifteen runs).

arm	mechanism it implements	faculty / model	read
predict-then-check (180B)	`predict(cells, note)` before each action; the sandbox reports HIT/MISS against the real diff	predictive coding	adopted: 474 hits, 140 misses, 32 unpredicted; levels 42 / 40 — null to negative
solver consolidation + harness replay (180B)	the model keeps one `solve_level()`; the harness runs it at each level start under a budget	consolidation, replay	46 / 47 levels; 2 of 45 replays cleared a level: the model's solvers are in-turn subroutines, not policies
act-every-turn (180B)	29% of turns execute no action; after one such turn the opener requires `action(...)`	habit / System 1	43 / 47 / 45 levels; tokens per level 21.2k / 19.8k / 20.5k vs 20.5k; turns

arm	mechanism it implements	faculty / model	read
			−26%: the same thinking in fewer turns
orienting census (180B)	the harness executes each simple action once at level start and tabulates the diffs	orienting reflex	run lost twice to platform mount failures; not read
96k context / 40k drain; 4096-token yield (180B)	cheaper prefix; fewer truncated turns	—	43 levels, throughput 64.7 (+1.4 sd); 50 / 44 levels — null
physics facts in the prompt (27B)	which body the keys move, its position, step, blocking colours, learned from the agent's own moves	perception	636 appearances in 16 games; 37 levels, in band
planner as tools `body()`, `goto()` (27B)	the learned rules execute a stated route in one turn	motor planning	reaches 5 of 9 rigid targets in the sandbox; 13 calls in 5 games in play, 0 reached; in band
planner as player, gated (27B)	the physics acts once 20 predictions verified, ≤1 in 10 wrong, 150 stalled actions	model-based control	35 vs 34 levels: one gained, none lost; without the gate, 34 vs 34
animation summary in the action result (27B)	the discarded intermediate frames, described	perception	730 appearances, 25/25 games; 40 levels, in band
world-model note backfilled from reasoning (27B)	the cross-turn note the model leaves empty on 33–47% of turns	working memory	empties 33% → 20%; offline 38 vs 32; board +0.1 over six draws — weakly positive
retention ledger; mandatory hypothesis-experiment-verdict protocol (27B)	cross-game memory; prescribed reasoning form	working memory; metacognition	engaged in 25/25 transcripts; 23 vs 25 levels; protocol followed 900+ times, levels 25 → 18
imitation fine-tunes (27B, two)	QLoRA on 6,512 human pairs; then on the harness's own rendered prompts	procedural learning	65.5% held-out next-action accuracy, serves verifiably; 17 vs 20 levels; 3 vs 6 and 17 vs 27
executable world model, model-authored / harness-driven (27B)	a verified transition-model API in the sandbox; a transition graph planned by the harness	model-based planning	never invoked in 25 games ("I should use the worldmodel tool … but first"); 19 either way; harness-driven null at the source (98.3% novel states)

None moved depth. The pattern is uniform across thirty arms, two chassis and four months: the model adopts every protocol it is offered — predicts, keeps solvers, follows the hypothesis form, reads the facts, calls the tools at level 1 — and reaches the same levels. Where a tool was used it reached the target the model chose in the optimal number of moves, and the target was wrong. Three arms were void before they were null, and their reasons are methods results worth more than the arms: injection through a method the harness defines and never calls; a setup that overwrites the environment the notebook set; a wrapper with a fixed signature that crashed under a bundle's extra keyword. Rule adopted: before reading a run's levels, grep the transcripts for the injected sentence and count helper *calls* in the model's code. A fourth lesson came from a table of closed levers: ours listed "KV cache" as closed after 24 and 48 GiB failed to boot under speculative decoding; a two-person team found that with the drafter off the same lever opens at 7 GiB, and their build drew 4.25 where ours drew 2.87. A row in a closed-lever table

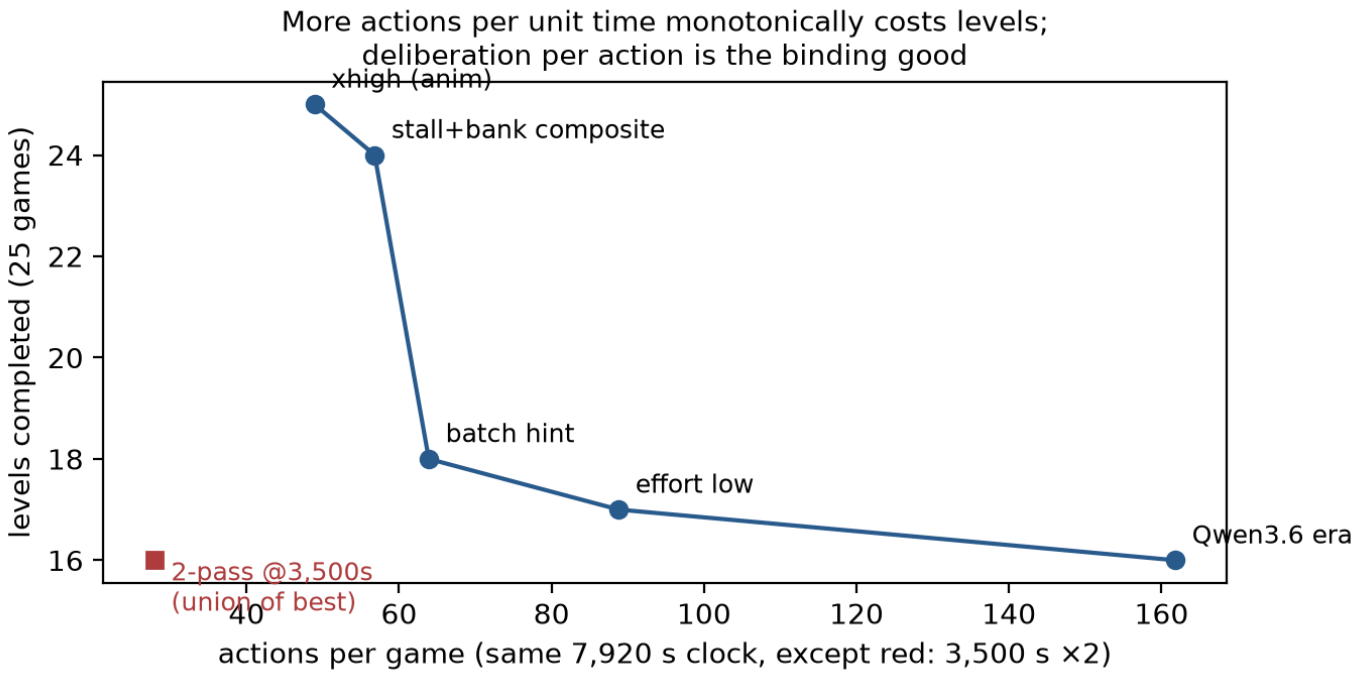
means closed under the condition measured, and the condition belongs in the row. One engine fact belongs here too: the harness advertises **UNDO** in every game, and on the local engine UNDO reverts the board in 1 of 25 games; in 17 it is a no-op that still costs an action (`scripts/test_undo_offline.py`). The model never used it, so the defect is latent, but any design that relies on reversal — our census was one — is void on this engine.

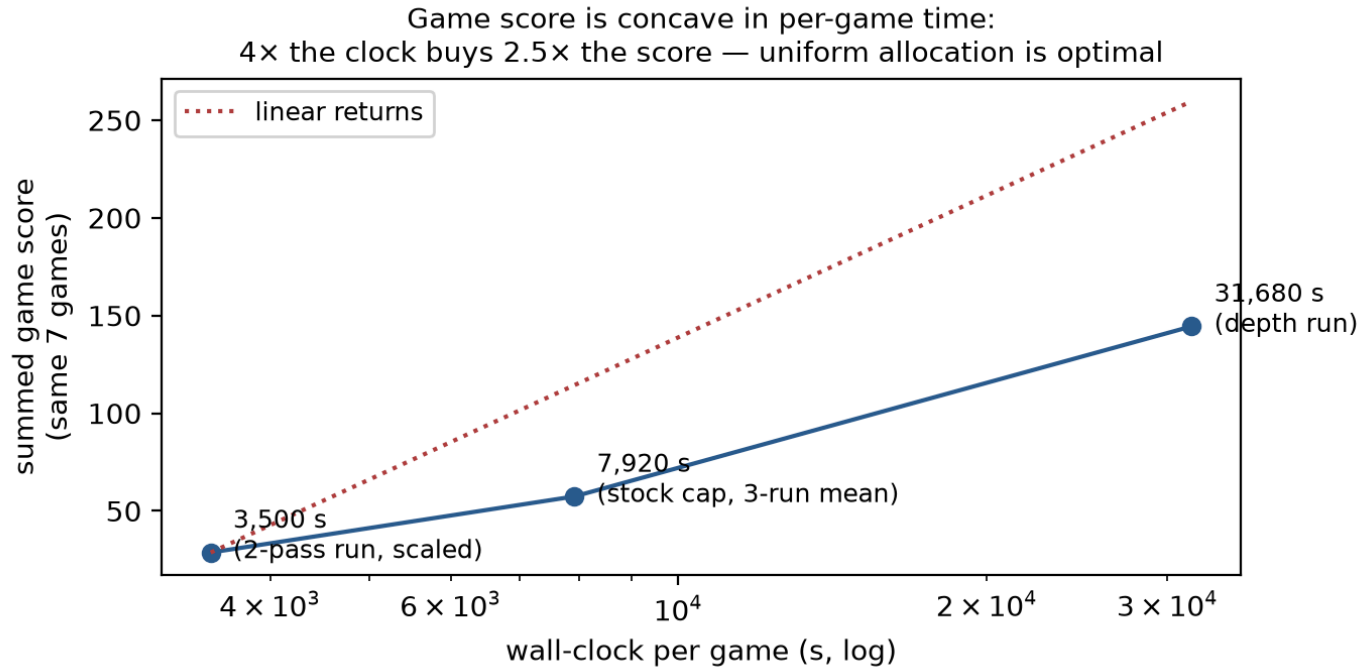
The one transferable positive is a gate, not a mechanism: the physics planner helped (35 vs 34, none lost) only when it had *earned* the budget — predictions verified on this level, a low error rate, the model stuck for 150 actions. A mechanism added to a competent model must earn its budget or stay out of the way; every ungated mechanism we measured took actions the model would have used better.

7. Results IV — resource economics: allocation and throughput

7.1 Time is concave; the queue buys actions, not levels

On the Milestone-1 chassis, every intervention that increased actions per unit time — a lower reasoning effort, batching hints, halving the per-game cap to buy a second pass — cost levels monotonically; giving seven games four times the clock returned 1.77× the levels and 2.51× the score, concave, so the stock uniform allocation was approximately optimal there. Removing that chassis's serving queue (KV cache 7 → 12 GiB; time to first token 66 → 25 s; 1.7× the actions) returned 44 levels against 45, 42, 42: the queue converts into actions, and actions do not convert into levels. Lossless speculative decoding moved levels 27 → 29, within noise, because at 25 concurrent games that workload was prefill-dominated and decode was not the resource the frontier was short of (Fig. 5). The largest single lever that summer was one nobody designed: upgrading the serving engine two minor versions moved 20 → 27 levels — larger than every deliberate modification — which bounds how much "agent design" can claim credit for score differences between entrants running different stacks. Two corollaries of the concavity (Fig. 6) survive the change of chassis: the games stuck at level 1 in every run stayed stuck with twice the time, and the per-game spread is wide (one game returned 9.0× for 4× time, another 1.0×), so adaptive reallocation has headroom where simple concentration does not — which is the question §7.2 answers with a simulator on the newer chassis.





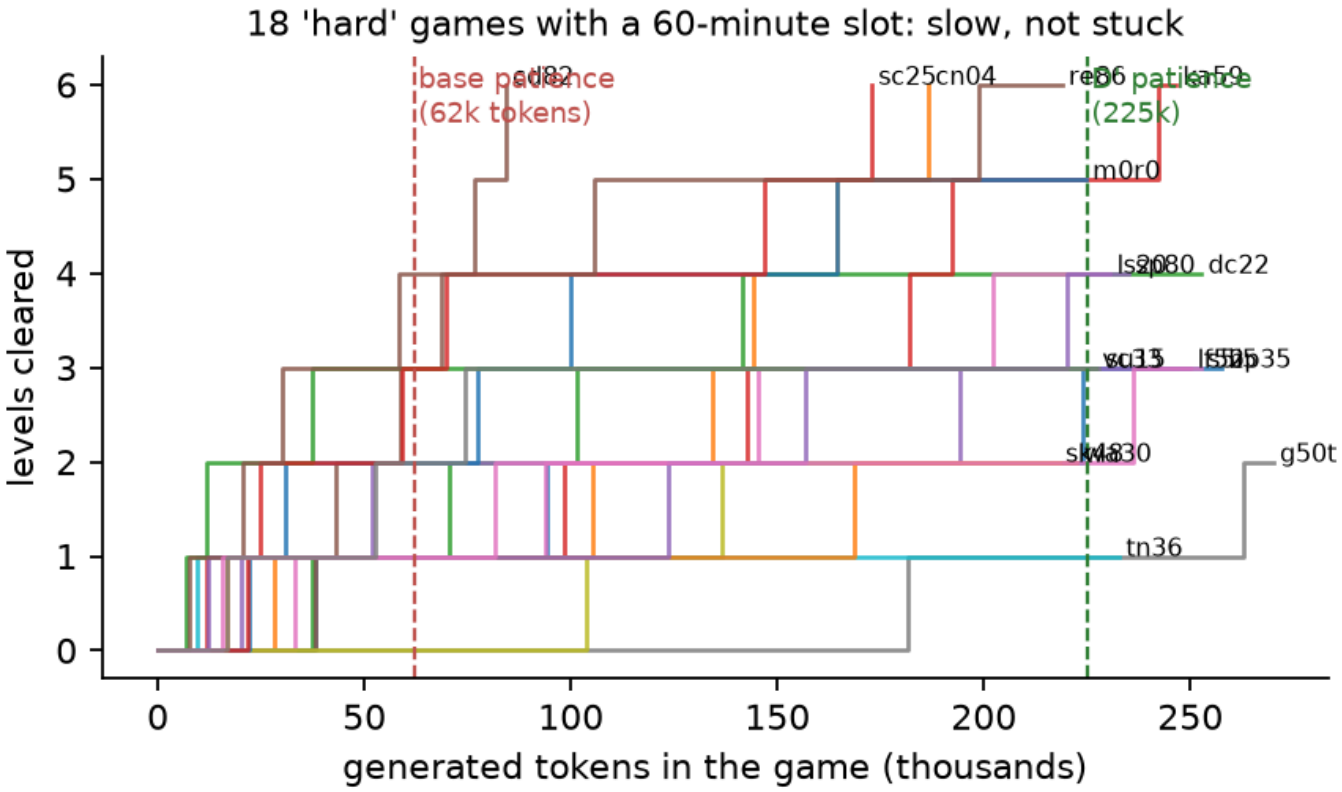
7.2 Allocation is the only lever that acts solely in the hidden run — and its ceiling

The contest run plays ~110 games on one GPU with ten model slots; a game holds a slot until its context is trimmed (58k generated tokens), then re-queues, and the highest-priority waiter takes the slot. Nothing on the ten-game demonstration can see this; the board can, and the field found it: a retuned priority ("D'", Affectify, 2026-10-04) on an otherwise unchanged notebook moved the author's 27–31 to a displayed 31.5.

The instrument. `scripts/sched_sim.py` is an event-driven simulator of that run. Its inputs are real trajectories — per action, the generated tokens and whether the action cleared a level — from 213 game-

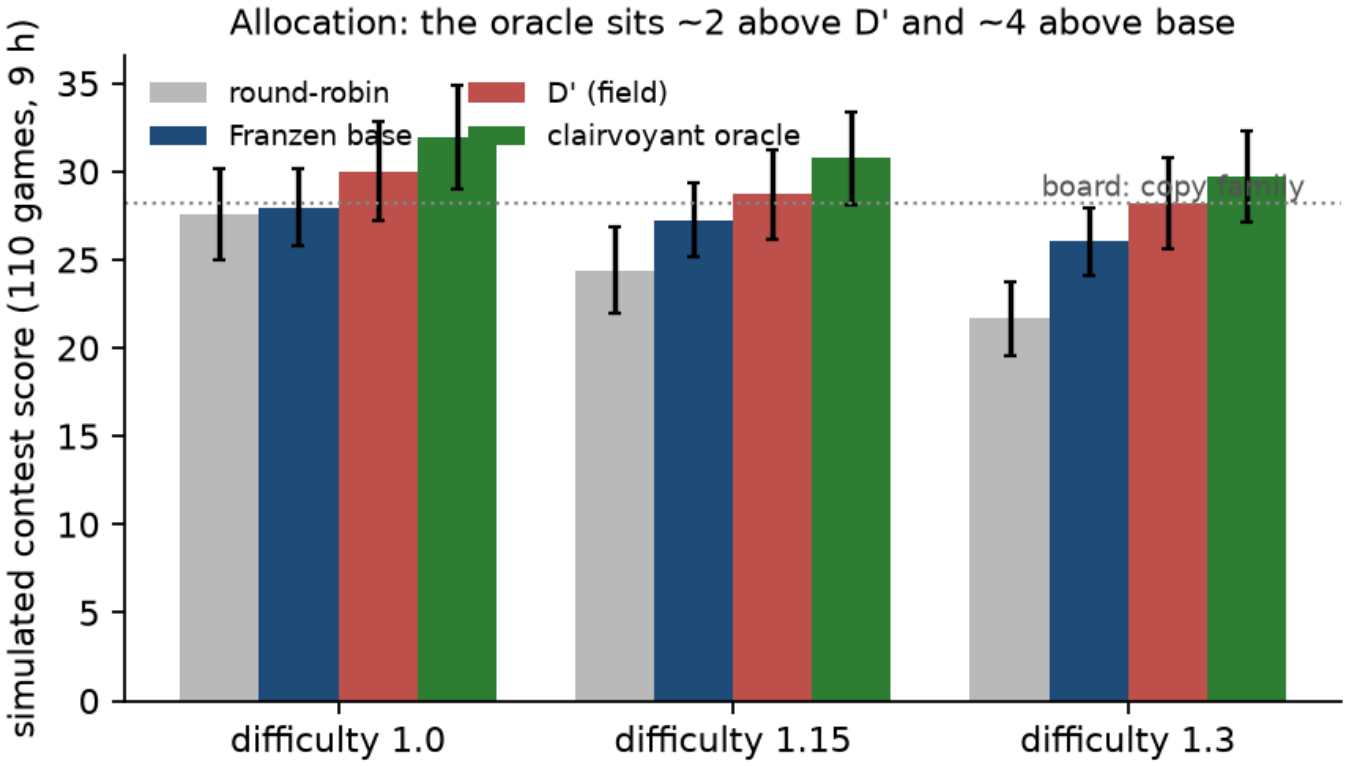
runs of the chassis, including the fifteen games the demonstration excludes, recorded with 60-minute slots. Its cost model is generation tokens divided by per-stream throughput, calibrated at 61 tokens/s (quartiles 58–64); the recorded per-action wall-clock is queue-inflated roughly thirty-fold and is never summed. It reproduces the gate (never-started games enter above every priced waiter unless the policy prices them; the queue is re-priced at every pump; the tail bonus fades). **Validation:** with no contention it reproduces each demonstration run's levels exactly (44, 50, 40); on a real 25-game, 10-slot, 25-minute run it gives 25 levels and 937 actions against the observed 25 and 938. **Calibration:** hidden games are harder than public ones (the author's public-25 mean was 46.5 against 27.9 semi-private), so each action's tokens are inflated by a difficulty factor; at 1.0–1.15 with 80% of the population drawn from the slow games the simulator gives the base priority 27–28 and D' 28.5–30, where the board puts them.

Why the retune works. The "hard" games are not stuck. Given a 60-minute slot each, eighteen of them cleared 69 levels (three fully); the median first clear came at 20k tokens and later clears 30–80k tokens apart, up to 180k (Fig. 7). The base priority discounts a level by a Gaussian in tokens spent with scale 62k, so it abandons these games mid-level; D' uses a linear 225k-token patience and keeps feeding them. Patience on slow games is the mechanism, not any insight about which games are promising.



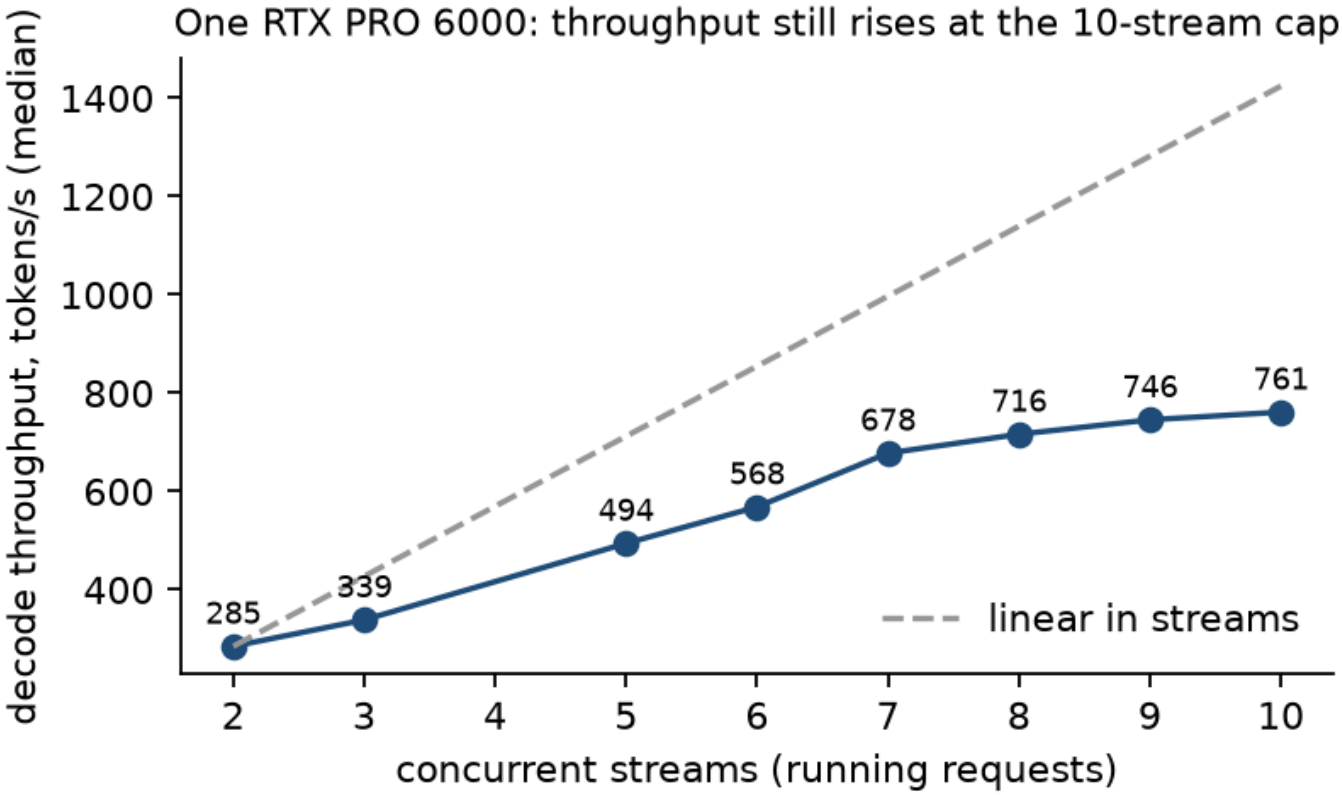
The ceiling. A clairvoyant policy that knows each game's tokens to its next clear bounds anything a priority formula can do. Across the calibrated regimes it scores 31.9 / 30.7 / 29.7 where D' scores 30.0 / 28.7 / 28.2 and the base 28.0 / 27.2 / 26.0; round-robin 27.5 / 24.4 / 21.7 (Fig. 8; eight seeds of 110 games). The lever is worth about +4 over the base and +2 over D', and D' drawn daily is worth about +1 per draw (§2.3). A searched policy of our own finished +0.4 to +0.9 above D' on held-out seeds, inside

that headroom and inside the draw noise. The reason the lever is small is total slot time: 110 games share 5,400 slot-minutes, 49 each, and the slow games need 60 to show what they can do.



7.3 Throughput binds

The server's decode throughput still rises almost linearly to the ten-stream cap — 285 → 761 tokens/s from two to ten streams (Fig. 9) — with a 94% prefix-cache hit rate, so prefill is not the cost; the MTP drafter's acceptance length is 3.1 of a maximum of 4, and the model's QSA attention rejects any deeper draft at graph capture, so speculative depth is closed. Twelve streams read 22.4 on the board over three draws against the copy's 26.3. What remains is the model's token cost per level — and §5 says what those tokens are spent on.



8. Theory: why, at three levels

Computational level. The task is joint inference of a game's dynamics and its objective under core-knowledge priors, from a few interactions, followed by efficient action. The human solves it in one or two interactions because both halves are drawn from small, structured hypothesis spaces: objects that cohere and persist, agents that act toward goals by efficient means, objectives that are compositions of a few relations (reach, collect, match, align). RHAE measures the efficiency half directly and the inference half through reach.

Algorithmic level. The ledger sorts the task's parts by what computes them. The local parts — which thing is the agent, how it moves, what blocks it, what colour ends a level — are learnable by counting from a handful of transitions (72%, 76%), and a planner over them reaches goals in fewer actions than humans. The global part — the objective's causal chain — is a structured search over compositions the counting theories do not contain. The model performs that search in context, slowly (six to ten turns per level), and succeeds on four games in twenty-two when asked to make the result explicit. Every mechanism we attached — prediction error, replay, forced acting, exploration, tools, memory — addresses a local part or a learning signal, and none addresses the search, which is why all read null: the model adopts them and the search is unchanged.

Implementational level. The brain pays for the search with dedicated systems and a lifetime of priors; the agent pays in generated tokens on one GPU inside nine hours. That makes allocation and throughput the whole economics of a run, and both are measurable and bounded: throughput scales to the stream cap and is capped by the model's attention, allocation by total slot time at +4. The economics

explain the field's shape — a copy band two points below its own display, a retune worth a point, leaders tens of standard deviations above — and they explain why the deficit cannot be bought back with harness: the thing that binds is the search, and the search is the model's.

9. Progress: what to build, with measured feasibility

The ledger prescribes a division of labour. Code owns what counters already predict — agent tracking, dynamics, blocking, the goal signature (72% of movement, 76% of completions) — and a planner owns motion, which it does in fewer actions than humans. Every token then goes to the one thing left: authoring the objective's causal chain and testing it against the engine, with the harness enforcing the gate §6 found — a mechanism earns its budget by verified prediction or stays out of the way. The simulator prices the prescription. Holding everything else fixed and scaling every action's token cost — the quantity a code-owned theory removes — the calibrated run moves as follows (base priority / D' / clairvoyant oracle; Appendix D.8):

tokens per action	**×1.0 (today)**	**×0.8**	**×0.7**	**×0.6**	**×0.5**
base priority	27.5	30.4	31.7	32.9	34.8
D'	29.2	30.7	32.3	33.6	35.5
oracle allocation	31.0	33.0	34.6	36.3	39.0

Direction keys are 58% of the agent's actions on this chassis and a counter predicts 72% of such moves; if the harness executes those without a model turn, and the model's deliberation shrinks in proportion, the token cost per level falls by 20–40% — worth +3 to +6 on the board at either allocation, more than the whole allocation lever, and the two compound (oracle at half cost: 39). The returns are sublinear in the simulator because the hidden games' objectives still cost what they cost; the projection is a bound on what cheaper motion buys, not on what inferring the objective would. Three falsifiable predictions follow. (i) Levels per token rise in proportion to the share of transitions the code theory predicts, because those are the tokens the model stops spending. (ii) A model that states the objective chain correctly after one completion clears the next level in human-like action counts; one that does not never does, however many actions it is given. (iii) No reallocation of slots lifts a Franzen-class chassis past ~32 on this board. Each is testable with the instruments in this paper and no GPU budget beyond a demonstration run.

10. Limitations

Our accuracy is low: our best public draw, 33.55, is a lucky draw of a 27-point vehicle, and we claim no leaderboard progress. The allocation ceiling is a calibrated simulator result whose absolute scale is fitted, not predicted; the ordering of policies was stable across every regime tried. The determinism result rests on 380 genuine repeats. Each arm used one open model per chassis; a model that reads its context differently may use what this one ignored. The theory space of §5 is the simplest one; a richer space would predict more of the residual third and still not the objective, which is the claim. The ledger's neural mappings are hypotheses that organise measurements; the rows marked "null / untested" and

the exploration row in particular have no evidence behind their mapping, and the hook tax has no mechanism. sc25 remains unexplained.

11. Conclusion

We set out to find where an LLM agent fails on a benchmark built from the priors humans solve it with, and we found it by dissection rather than by building. The agent has the agent signatures Spelke names — contingency and efficiency — at or above human level; the observation it reasons from can be made a sufficient state at no cost; the dynamics of a game and the colour that ends a level are learnable by counting; and the slot allocation of the hidden run, the one lever that acts only there, is bounded at four points. What is missing is the inference of a level's objective as a multi-step causal chain from one or two interactions. A counter cannot supply it, a search cannot, six learning mechanisms attached to the harness did not, and the model supplies it slowly, in tokens, and explicitly on four games in twenty-two. The leaderboard that is supposed to measure all this displays each team's best draw and moves by copying. The contribution is the ledger and the instruments that filled it — a noise model, a calibrated simulator, a theory-induction test, a goal signature, a planner loop, thirty arms with a script each — and a prescription that follows from them: let code own what counters predict, let a planner own motion, and spend every token on the chain. Three predictions say how to tell whether that is right. We did not solve the benchmark; we located the faculty that does.

Appendix A. The published scoring formula is not the one that scores you

The technical report gives the level score as `S = min(1.0, h/a)^2` and states that it caps "the per-level efficiency an AI can receive at 1.0x human baseline". Under that formula, beating the human baseline is worth nothing.

The gateway that scores submissions does something else:

```
# Calculate score as ((baseline_actions / actions_taken)^2 * 100) max 100
if actions_taken > 0:
    score = ((baseline_actions / actions_taken) ** 2) * 100
    score = min(score, 115.0)  # Cap at 115
```

The comment says 100 and the line below it caps at 115; the specification agrees with the comment. The order differs too — the paper caps the ratio then squares, the code squares then caps the score — so about **6.8% fewer actions than the second-best human earns 15% of headroom the published formula says does not exist**. We have not established whether this is a known erratum, and do not present it as an error.

Appendix B. Reproducibility: every claim to its script and artifact

All code we authored is MIT-0 in the public repository (`tantan0327/arc-agi-3-agent`); the research scripts below join it on publication. Third-party components keep their own licences (SOLUTION.md lists them). Every leaderboard number is a row in `SUBMISSIONS.md` with its date and kernel version; every offline number comes from a run's `benchmark.json` and `intermediate_states.pkl`, read with `scripts/load_recorded_states.py`.

claim (section)	script / artifact	ground truth
hook tax 0.448; null control 0.60/0.59 (§2.2)	`SUBMISSIONS.md` rows 2026-08-06–08-17 (five hook modifications, the null control and its second draw, the empty-hook and setup-route controls); the null-control kernel diff is one import and one print	leaderboard
max-statistic: sd 0.10–0.20, 67 expected vs 66 observed, as-is fork 1.22 vs 2.23, third family 2.45→3.92 (§2.3)	`SUBMISSIONS.md` (draw counts per configuration); the binomial arithmetic is stated in the text; leaderboard snapshots dated in the text	leaderboard
human replays 22/22 in sync, 6,512 actions (§2.1)	`scripts/replay_humans.py` over `official_human_replays/*.json` against the vendored engine	engine
random floor: 2/25 start, 0/25 finish (§2.1, §4.1)	`scripts/evaluate.py --agent <framework random template>` with the budget rule in §2.1	engine
elimination table rows (Appendix E; §5.4, §6)	each row names its run; runs are Phase A kernels built by `scripts/build_duck_notebook.py` (`SETUP_SUB=` names the arm) or `scripts/build_bundle_fork_v2.py`; levels read from the run's `benchmark.json`	engine / leaderboard
98.3% novel states, 0.9% rediscoverable (§4.5)	`scripts/measure_search_vs_perception.py`	human replays
starts 11/25, per-level action counts, stall signature (§4.1)	`scripts/spend.py`, `scripts/diagnose.py` over a run's artifacts; entropy signal `scripts/measure_progress_signal.py`; state-novelty slope `scripts/measure_state_novelty.py <run_dir>`	engine + human replays
throughput frontier, time concavity (Figs. 5–6, §4.1)	the six runs' `benchmark.json` (kernel versions in `SUBMISSIONS.md`/TASKS.md); plotted ad hoc — the plotting code is not in the repository, the numbers are	engine
determinism ladder 43.2 → 76.3 → 90.0 → 99.1 (§4.3, Fig. 2)	`scripts/test_worldmodel_kit.py` reproduces the ladder on recorded data; the kit is `duck_delta/worldmodel_kit.py`	engine (recorded runs)
abstraction ladder 98.3 → 37.9 (§4.5, Fig. 3 left)	`scripts/measure_abstraction_ladder.py`	human replays
scorer: policy vs dynamics views, plan-phase 53%/42% (§4.5)	`scripts/score_abstraction.py` (built-in rungs; `--module` for any abstraction)	human replays
hand-written references 100 / 95 / delta 79 (§4.5, Fig. 3 right)	`scripts/gpb_reference_abstractions.py` scored by `scripts/score_abstraction.py --module`	human replays
replay-only planner: 70/70 levels, −24% actions (§4.5)	`scripts/test_v10_planner.py`	human replays

claim (section)	script / artifact	ground truth
agency classification per game (§4.4)	`scripts/measure_agency.py <run_dir>`; raw tables `docs/paper/agency_runs.txt`	engine (recorded runs)
authoring probe 4/22 over 437 authorings (§5.3, Fig. 4 left)	`scripts/gpc_author.py` (loop; `--mock` replays the references); per-run results `scratchpad/gpc/gpc_v{3,4,5,7}_results.json`; grades recomputed by `scripts/make_paper_figures_v2.py`	human replays
closed loop 21 vs 27/28 (§5.3, Fig. 4 right)	`duck_delta/v10_explorer.py` shipped by `scripts/build_bundle_fork_v2.py`, arm `q38_nightly_v10`; artifacts `scratchpad/v10/`; offline engine test `scripts/test_v10_explorer.py`	engine
scoring formula cap 1.15 (Appendix A)	the gateway's scorecard source, quoted; `scripts/action_gate.py` applies the same arithmetic offline	—
board noise (14 draws mean 2.87 sd 0.47; B81 nine draws mean 4.25) and offline band (32–41; 45/42/42) (§2.4)	`SUBMISSIONS.md` rows 2026-09-04–09-29; run dirs in `TASKS.md` (9/16–9/28 sections); `scripts/compare_probes.py`	board / engine
queue removal: KV 7 → 12 GiB, running 10 → 17, TTFT 66 → 25 s, actions ×1.7, levels in band (§7.1)	`scripts/build_kv_variant.py 12`; runs `arc3-thui-b81-asis` v1, `arc3-b81-rep` v1–v2, `arc3-b81-kv12` v1 (`benchmark.json`, `vllm-openai-server.log`)	engine
efficiency pricing +10–15%; "every game to level 3" = 21.0 (§7.1)	`actions_per_level` / `base_actions_per_level` in `benchmark.json`; level counts from the human replays	engine / replays
note empties 33% → 20% and 47% → 27%; board +0.1 (§6, Appendix F)	`scripts/patches/note_from_reasoning.py`; `scripts/build_probe.py notefull`; `scripts/build_stack_on_b81.py cand-c`; transcripts of runs `arc3-probe-notefull`, `arc3-cand-c`	engine / board
animation channel 730 appearances, facts 636 appearances, both null (§6, Appendix F)	`scripts/patches/animation_channel.py`, `scripts/build_facts_patch.py`; runs `arc3-probe-animfull` v2, `arc3-probe-factsfull` v2	engine
goto tools: 5/9 rigid targets in the real sandbox; 13 calls / 0 reached in play (§6, Appendix F)	`duck_delta/goto_tool.py`, `scripts/build_goto_patch.py`, `scripts/test_goto_tool.py`, `scripts/test_goto_refused_edge.py`; runs `arc3-probe-gotofull`, `arc3-b81-goto` v2	engine

Two honesty notes. The offline instrument has a standard deviation of 0.91 score points against the board's 0.10 (§10), so offline level counts are reported as counts and compared only within a run family sharing an engine and a clock. And the two earliest figures were plotted outside the repository; every number on them is in the run artifacts named above.

October 2026 additions. `scripts/sched_sim.py` (simulator; `--validate`, `--calib`, `--compare`), `scripts/sched_search.py` (policy search), `scripts/patches/priority_form.py` (the pluggable priority, forms base/dprime/ours; `scripts/test_priority_form_stub.py`), `scripts/build_hardtrace.py` (trace-

collection notebooks), `scratchpad/sched/traces.json` (213 trajectories), `scripts/test_undo_offline.py` (UNDO on 25 games), `scripts/test_theory_induction.py` (theory induction by code on the human replays; `docs/results/theory_induction_v3_2026-10-21.txt`), `scripts/test_theory_planner.py` (the planner loop; `docs/results/theory_planner_2026-10-21.txt`), `scripts/diagnose_theory_failures.py`, `scripts/patches/{orienting_census,act_rule,pred_error}.py` with their stub tests, `scripts/make_paper_figures_v3.py` (the four October figures), `docs/results/ scheduler_sim_2026-10-07.md`, `docs/results/scheduler_policy_2026-10-10.md`.

Appendix C. Leaderboard and noise statistics in full

Two fixed configurations at the 1-point scale, four draws each. Unmodified harness 0.98, 1.11, 1.14, 1.23 (mean 1.115, sd 0.103); Qwen3.8 family 1.87, 1.65, 1.43, 1.50 (mean 1.61, sd 0.195). Under the measured family distribution one draw reaches 2.0 with probability 0.023; if ~600 teams keep their best of about five draws, P(best of 5 ≥ 2.0) = 11.1% and the expected number of teams ≥ 2.0 is 67; 66 were observed on 2026-08-22. The top public kernel (advertised 2.23, "rank 9") publishes an offline benchmark that completes 25 of 183 levels — our configuration's result under the same harness, games and clock; forked byte for byte and submitted, it scored 1.22 (2026-08-26).

Third family. Six draws of one unchanged Qwen3.8-Flash-Next recipe: 2.45, 2.75, 2.80, 3.01, 3.16, 3.92 (~200th → ~77th place on the same bytes). Fourteen draws of the same recipe between 2026-09-04 and 09-18: mean 2.87, sd 0.47, range 2.14–3.92; with the note backfill (§6), six draws, mean 2.96. A byte-identical copy of Thuitanium's B81 build on our account: 5.10, 3.62, 4.49, 4.10, 3.92 (mean 4.23, sd 0.55); the authors' own four draws 4.50, 3.86, 3.03, 5.36; nine draws across two accounts mean 4.25, sd 0.7.

The copy wave. On 2026-10-03, 76 teams with at most two submissions since 10/1 scored mean 26.8, sd 3.55, median 27.3, 90th percentile 30.9, 99th 32.7, maximum 36.1; the author's best over 87 submissions was 27.9; our copy's first draw 24.65. A bootstrap of that distribution puts the expected maximum of thirty daily draws at 33.4. Our copy draws (scored): 24.65, 26.26, 26.88, 28.27, 25.82 (mean 26.4). Our D' draws (scored): 23.95, 23.66, 28.83, 33.55, 29.18, 26.23, 24.88, 25.55, 29.35 (mean 27.2); three further reruns (10/10, 10/17, 10/21) failed at dataset mount and scored 0.00. The copy band on 2026-10-20: 1,661 teams between 20 and 34, mean 28.7, sd 2.8; the top-5 line 49.85, #10 43.94,

20 37.9, #50 35.0.

Offline instrument. Twelve full-length runs of one 27B vehicle: 32–41 levels (sd ≈ 3), score 6.0–9.4 (sd ≈ 1); three as-is runs of the B81 chassis 45, 42, 42 levels; the Milestone-2 ten-game demonstration band 44, 50, 40 levels (turns 367/372/364). Per game: lp85 2–6, ar25 2–5, ft09 2–4, tr87 0–3 across three runs of one notebook. Independent report in the forum (Kaggle discussion 743723, 2026-09-27): two runs of an identical build, 2.60 and 1.45.

Appendix D. Per-game tables

D.1 Determinism residue (§4.3).

game	repeats	contradictions	in animation	off-screen	agree
lf52	52	49	49	0	6%
sc25	41	36	1	35	12%
bp35	34	5	2	3	85%
15 others		0			100%

D.2 Abstraction ladder on the 6,512 human decisions (§4.5).

representation	novel state	known state+action
A0 masked board	98.3%	0.9%
A1 object set (colour, count)	37.9%	49.0%
A2 object layout (positions / 4)	92.1%	5.1%
A3 colour census + avatar position	89.4%	7.4%

Hand-written abstractions: tr87 (cursor column + tile band) determinism 100%; m0r0 (wall mask + avatar) 95%; s5i5 (bar sizes and anchors) per-control delta consistency 79%; revisit rates 10–27%; the replay-only planner re-solves 70/70 level segments in 24% fewer actions than the humans (48% maze, 26% tile, 5% progression). At an identical masked board, humans repeat their modal action 53% of the time; at an identical maze position 42% (plan-phase is real and not in the state).

D.3 Agency classification (§4.4), stable across three harness generations. One object accounts for ≥80% of translating moves: 3–5 of 17 games (e.g. dc22 97%, bp35 72%, g50t 54–62%); partially (10–76%): 6–8 games; never or once: cn04, tu93, lf52, cd82, re86, ls20 (0% in every run), ar25 ≤1%. Raw tables in `docs/paper/agency_runs.txt`.

D.4 Goal-colour census (§5.2). Completions whose entered/clicked colour is the game's modal colour: dc22 6/6, g50t 7/7, ka59 7/7, r11l 6/6, sk48 7/7, tn36 7/7, vc33 7/7, sc25 5/6, re86 6/8, ar25 5/7, cn04 4/6, su15 6/9, bp35 5/8, lf52 6/10, m0r0 3/5, tu93 4/7, wa30 1/4; sb26, sp80, tr87 no tracked goal cell. Total 98 of 129 (76%).

D.5 Theory-based planner (§5.2), levels 2+ in the real engine. ar25 0/2, bp35 0/1, cn04 0/1, dc22 0/2 (L2: 20 actions, human 55), g50t 0/6, ka59 0/6, m0r0 0/5, re86 0/1, sc25 0/5, tu93 0/8; 0 of 37 overall. Click-only games and games whose level-1 completion was not a direction action were not tested.

D.6 Sixty-minute traces (§7.2). Levels cleared / levels, with cumulative tokens (thousands) at each clear: bp35 3/9 [38, 94, 224]; cn04 6/6 [28, 105, 134, 144, 164, 187]; dc22 4/6 [37, 71, 102, 142]; ka59 6/7 [12, 98, 143, 182, 192, 242]; lf52 3/10 [20, 52, 194]; cd82 6/6 [38, 43, 59, 69, 77, 84]; ls20 4/7 [33, 94, 145, 202]; g50t 2/7 [181, 263]; sk48 2/8 [104, 136]; tn36 1/7 [9]; m0r0 5/6 [22, 31, 77, 100, 165]; wa30 2/9 [17, 169]; vc33 3/7 [7, 12, 37]; sc25 6/6 [22, 25, 59, 70, 147, 173]; sp80 4/6 [12, 124, 157, 220]; re86 6/8 [8, 20, 30, 58, 106, 199]; s5i5 3/8 [15, 82, 236]; su15 3/9 [17, 53, 74]. Tokens to the first clear: median 20k, maximum 181k; 2 of 18 above 62k.

D.7b Token-cost projection (§9). Calibrated simulator, hard_frac 0.8, hazard 0.1, seeds 40–49, 110 games, nine hours, ten slots; difficulty 1.10 is today's cost. Token cost ×1.0 / 0.9 / 0.8 / 0.7 / 0.6 / 0.5:

base 27.5 / 29.6 / 30.4 / 31.7 / 32.9 / 34.8; D' 29.2 / 30.1 / 30.7 / 32.3 / 33.6 / 35.5; oracle 31.0 / 31.9 / 33.0 / 34.6 / 36.3 / 39.0 (sd across seeds ≈ 3). Agent action mix on the chassis (31,562 actions): direction 58%, click 38%, other 5%.

D.7 Serving (§7.3). Decode throughput by running requests (median tokens/s): 2: 285, 3: 339, 5: 494, 6: 568, 7: 678, 8: 716, 9: 746, 10: 761. Prefill batches 640: uncached 2.0M tokens, cached 32.8M (hit 94%). MTP accept length 3.1, accept rate 0.70. KV usage 3%, Mamba cache 36 of 60 slots. Per-stream throughput at the game level across fifteen demonstration runs: median 62.6 tokens/s, sd 1.5.

Appendix E. The elimination table on the Milestone-1 chassis (§5.4, §6)

against one of those two ground truths.

hypothesis	verdict	evidence
inhibition of return	refuted	humans revisit 26% of clicks, median gap 1
the agent random-walks	refuted	true cycling 2.3%, uncorrelated with scoring
contingency detection	near parity	94% of agent actions change the board, human 98.7%
object identity as goal vocabulary	monotonically worse	colour 13/22, size-class 12/22, exact shape 3/22
cross-game forgetting is the gap	refuted	retention + a factual evidence ledger engaged in 25/25 transcripts (goal model carried up to 68 turns vs stock 6.9%); levels 23 vs control 25
prescribing the form of reasoning helps	refuted	a mandatory hypothesis/experiment/verdict protocol was followed (900+ instances of each element across 1,255 actions) and levels fell 25 → 18 at unchanged action count
wasted actions are the gap	not the gap	13.9% of agent actions leave the board unchanged against a human 0.6%, but removing them does not follow through to levels
imitating the humans directly transfers	refuted in its natural form	a QLoRA adapter trained on all 6,512 human pairs reaches held-out next-action accuracy of 65.5% exact / 74.0% by type (offline baselines 30.4% / 68.2%), verifiably serves every request (requests route through the adapter's model id, which would 404 unregistered) — and completes 17 levels against its budget-matched control's 20. Imitation that measures clean offline does not survive the change of rendering between training context and play-time prompt
decode speed is the binding resource	refuted	lossless speculative decoding (a distilled draft verified by the target; the output distribution is preserved by construction) runs with healthy acceptance — mean length ~5 of 8 drafted, per-position 0.88/0.76/0.64… — and moves levels 27 → 29, inside single-run noise. The workload is prefill-dominated at 25 concurrent games; a 5× decode gain does not reach wall-clock
harness-matched	refuted (the confound	a second QLoRA trained on the *harness's own* rendered prompts and tool-call targets (97.3% of generations parse as valid tool calls) completes 3 held-out levels against the

hypothesis	verdict	evidence
imitation transfers	removed)	engine-only control's 6, and 17 of the public 25 against 27. Removing round 1's rendering gap did not rescue it: next-action imitation captures what a human did, not why it makes progress
retained context is the gap	not supported	a 100-turn history window on our fork completes 20 levels, inside the fork's own 18–21 baseline band (the 27-level control ran a different bundle; see §6.1) — retention bought nothing, the same drowning-in-context null as OpenAI's un-adapted harness, at the opposite scale
an executable world model, LLM-authored	model-limited, not method-limited	shipping a verified transition-model API into the sandbox (offline unit tests: the verifier absorbs 3/3 raw disagreements; the planner re-solves 85/85 recorded levels) the 27B never once invoked across 25 games, even when the contract was reworded to name the world model the most efficient path -- one transcript reads "I realize I should use the worldmodel tool to plan this. But first I need to understand R's movement. Let me gather data", then reverts to reactive play and never returns; 19 levels either way. It knows the better move and does not take it -- an agency limit, not a harness one. The published method that reaches ~99% used a frontier model
an executable world model, harness-driven	refuted at the source	if the harness built the transition graph itself and planned over it, no LLM needed — but **98.3% of human decisions land on a masked board state never seen before in the game** (0.9% rediscoverable). At the board representation there is no revisitable structure to plan over (§5.1 shows where there is, and what it costs to find it); this is why mechanical frontier navigation measured null (22 vs 25)

The last five rows share a spine, and together they are this paper's strongest negative result. Imitation of what the human did (twice, the second time with the rendering confound removed), retention of context, and an executable world model in both of its forms — LLM-authored and harness-driven — all fail, and

Appendix F. The 27B observation-pipeline arms in detail (§6)

the human replays and the game sources located three defects in what the agent is shown — not in what it reasons about:

1. **Intermediate frames are discarded.** An action animates over several frames; the harness keeps the last (`raw.frame[-1]`). On bp35 a seven-key batch animates over five frames with 1,038 transient pixels the model never sees; the game's mechanic is visible only there.
2. **The carried world-model note is empty when the model writes no prose.** The harness refreshes its cross-turn note only from assistant text; Qwen3.8-Flash-Next thinks and then calls the tool directly. On the as-is fork the note is empty on 33% of turns and on whole games (bp35 55/55 turns, sk48 53/53); on the animation-aware chassis, 47% (bp35 45/45, tu93 38/38, ar25 47/49). The labelled lines the system prompt asks for do appear inside the retained reasoning.
3. **The agent's own perception code is slow and wrong.** g50t: five of nine level-2 turns spent confusing the player with the goal bracket and estimating the cell pitch as 6.125; ls20: thirteen moves into walls and a level lost to an unplanned budget; m0r0: the right route found at step 21, the clock at step 22, tool timeouts from re-parsing every history frame.

Each defect got a fix, delivered as a monkey-patch cell after the bundle's setup, and each was measured at full length against the vehicle's band (§2.4):

fix	delivered?	result
note backfilled from retained reasoning when prose is empty	yes: empties 33% → 20% (as-is), 47% → 27% (B81); bp35 0 → 31 prose-less turns carrying a note	as-is: 38 levels against 32 in its one run, three diagnosed games cleared for the first time; **board +0.1 over six draws** (2.96 vs 2.87). On B81: 45 levels = 45. Weakly positive at best
animation summary in the action result and one prompt sentence	yes: 730 appearances in user prompts, 25/25 games	40 levels, every game inside the band — null
mechanical facts (which body the keys move, its position, step, blocking colours) learned from the agent's own moves and written into the prompt	yes: 636 appearances, 16/25 games (those with a rigid mover)	37 levels, inside the band — null
the same planner as callable helpers inside the python tool (`body()`, `goto(row, col)`)	yes; in the real sandbox against the engine it reaches 5 of 9 rigid targets at human action counts	in play: 13 calls in 5–6 games, **0 reached**, levels inside the band — on both chassis
the planner acting on its own once its fit is trusted	gate never earned in the real action budget	null

The model read "that action animated over five frames — the effect is visible only in the intermediate frames", read the measured facts about its own body, was handed a tool that executes a stated plan in one turn, and did not act differently on the diagnosed games. The diagnosis of what the agent could not see was right; the inference that seeing it would change the play was wrong for this model. Where the tool was used at level 1 it reached the target the model chose in the optimal number of moves — and the target was not what the level wanted. The failure categories that remain (wrong goal hypothesis on lf52 and sk48, wrong rule on wa30, the click-gated levels the rigid planner cannot represent) are inference, and no change to the observation pipeline reached them.

Three of the fifteen arms were void before they were null, and the reasons are the methods result of this section:

- **Dead injection points.** The first animation and facts runs injected their text through a method that the agent module defines and never calls (`_describe_last_outcome`); the user prompt is rendered elsewhere (`_build_user_prompt`). Both runs completed with clean logs and measured nothing. Rule adopted: before reading a run's levels, grep the transcripts for the injected sentence; count helper *calls* in the model's tool code, not injections.
- **Setup overwrites the environment.** The bundle's setup persists its analyzer settings into `os.environ` after the notebook's own cell has set them, so three early environment-variable arms measured the stock value. Overrides go in a cell inserted after setup.
- **Wrappers with fixed signatures.** The animation-aware bundle passes one more keyword (`animation_handler`) to the sandbox than the June bundle; a wrapper that named its arguments crashed all 25 games in nineteen seconds. The stub check had verified that the hook installed, not that a call went through.

And one result about tables of closed levers. Ours listed "KV cache" as closed after 24 and 48 GiB failed to boot — under MTP on. A two-person team found that MTP off opens the same lever at 7 GiB, and their build draws 4.25 against our 2.87. A row in a closed-lever table means *closed under the condition measured*; the condition belongs in the row.